IoT based Water Monitoring System

by
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CPE162P-4_E01
EMBEDDED SYSTEM WITH IOT

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Objective

Objective Overview:

- **To demonstrate how an IoT based Water monitoring system works.**
- **To be able apply what you have learned in logic circuits and switching theory, microprocessors, CPE160P, and CPE161P.**
- **To be able to apply embedded system with IoT using Arduino kit**
- **To be able to use Thingspeak for the data collection and analysis.**

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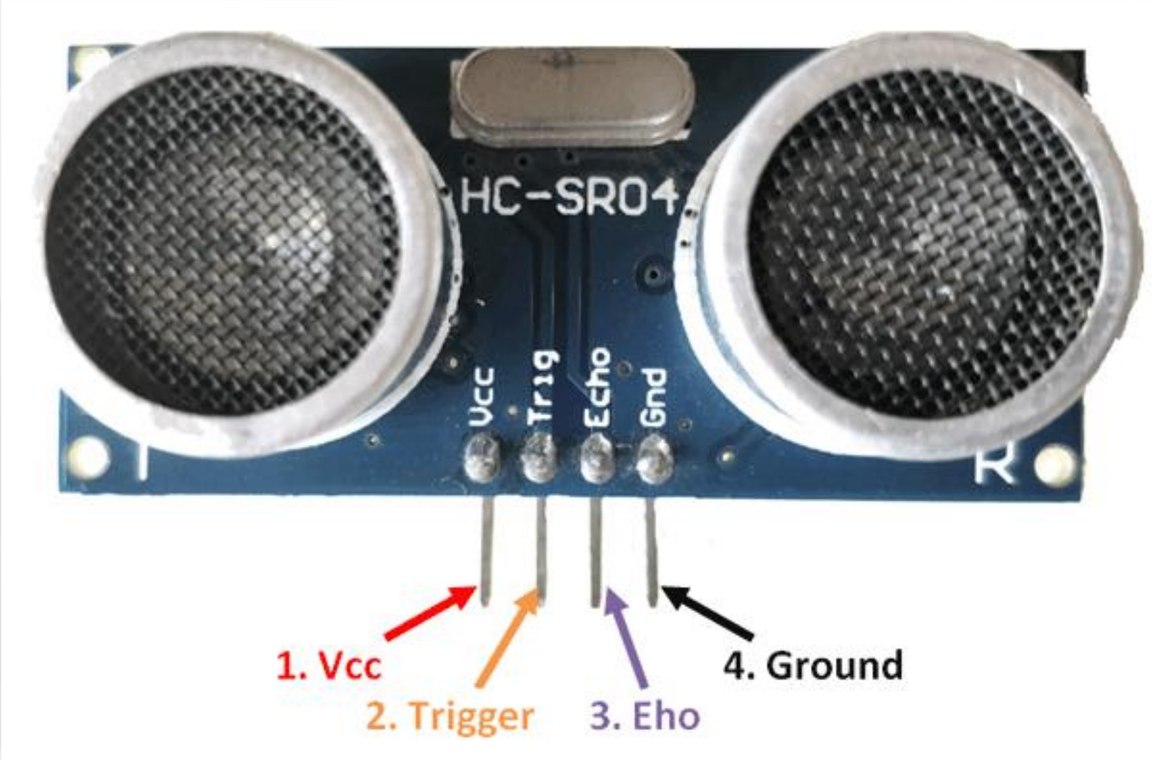
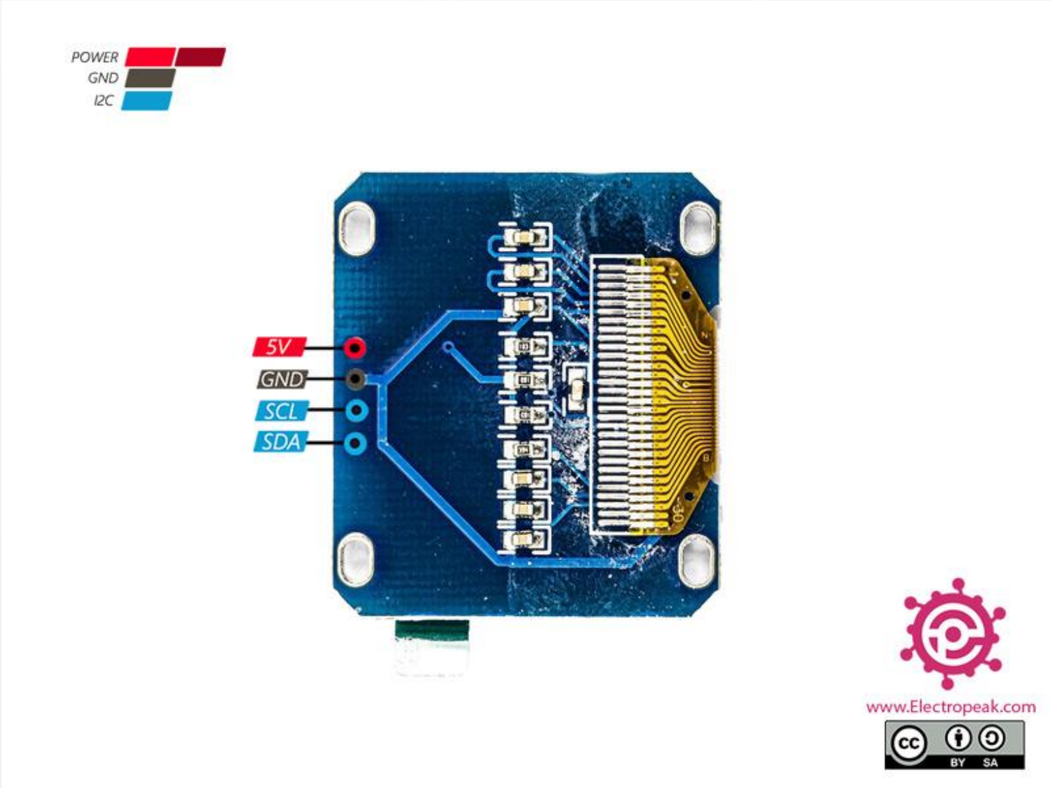
Materials



SOIL MOISTURE SENSOR



OLED DISPLAY

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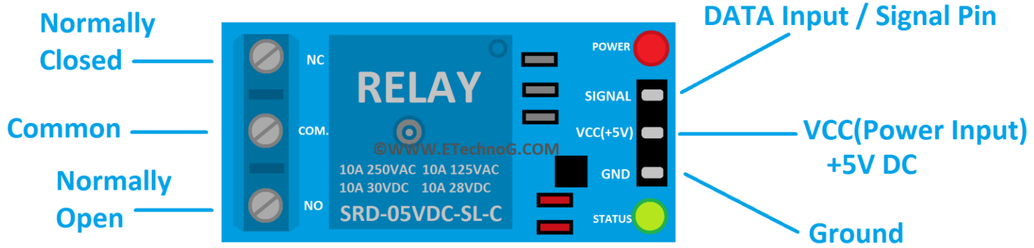
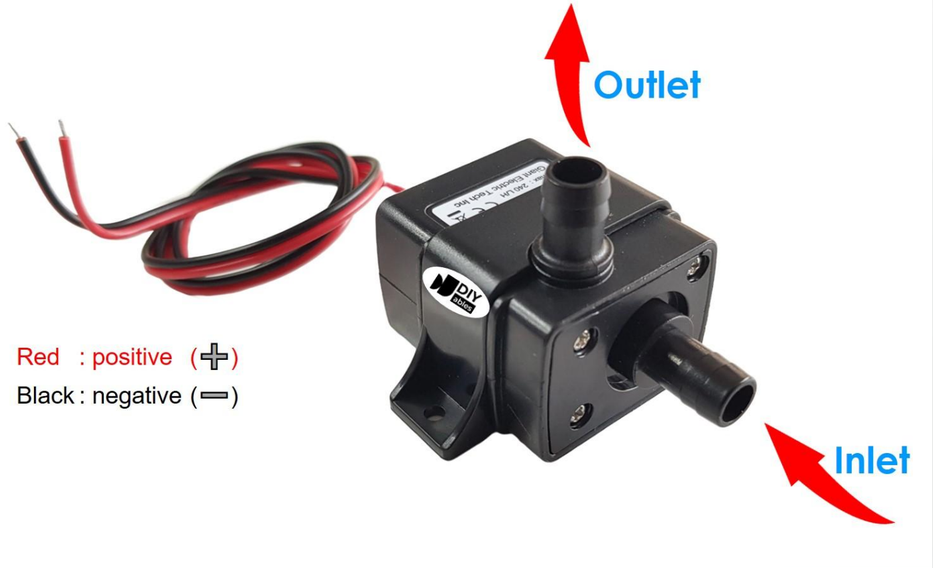
Materials

Figure.01: 5V Single Channel Relay Module Pinout

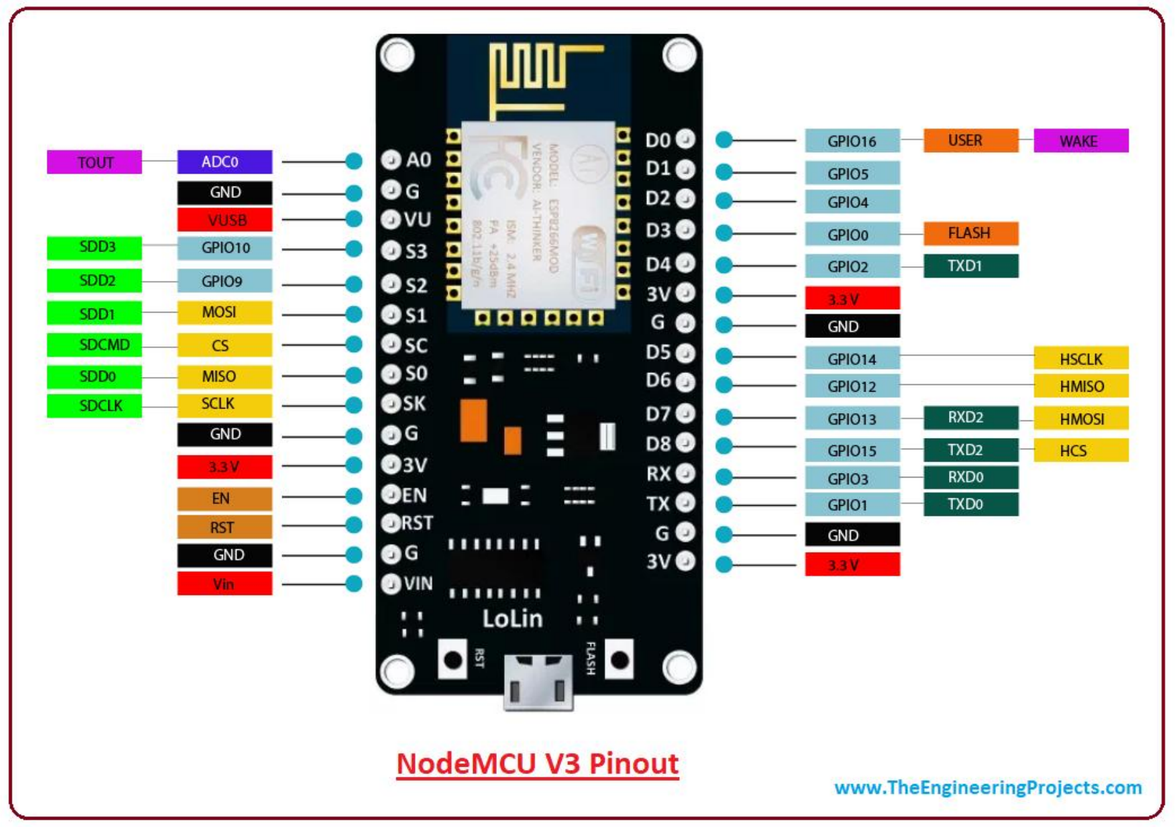




Relay

Water Pump

Materials



ESP8266 NodeMCU v3

Introduction

This experiment explores the integration of embedded systems and Internet of Things (IoT) technologies to develop an automated water volume monitoring system. Using an ESP8266 microcontroller, ultrasonic sensor, relay-controlled water pump, and OLED display, the system measures the real-time water height inside a cylindrical container and converts it into volume. The relay module simulates water flow by cycling the pump ON and OFF at various intervals, while sensor data is transmitted to the cloud via ThingSpeak for remote monitoring and analysis.

Key Concepts Covered

- Real-time water level detection and volume calculation using ultrasonic sensing
- Time-based pump control using relay logic and switching theory
- OLED-based Idisplay integration
- Wireless data logging and visualization using ThingSpeak

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PROCEDURE

(Software, Hardware, Circuitry)

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Software – Initialization

```
#include <ThingSpeak.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <splash.h>
#include "thingProperties.h"

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

#define TRIG_PIN 14
#define ECHO_PIN 12
#define PUMP_PIN 13

#define SDA_PIN 4
#define SCL_PIN 5

#define CYLINDER_DIAMETER 14.0     // in cm
#define MAX_HEIGHT 19.0            // in cm, max sensor distance when empty

unsigned long pumpOnDuration = xxxx;      // Change second for different water rate
unsigned long pumpOffDuration = 20000;    // 20 seconds OFF
unsigned long lastPumpToggleTime = 0;
bool pumpIsOn = false;
```

Software – Initialization

```
// ThingSpeak settings
WiFiClient client;
unsigned long myChannelNumber = 2980959;
const char *myWriteAPIKey = "CJP46XT2YT8S5SL2";

unsigned long lastUpdate = 0;
const unsigned long updateInterval = 15000;

float measureDistance() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH, 50000);  // Increased timeout
  float distance = duration * 0.0343 / 2;
  return distance;
}

```

Software – Setup

```
void setup() {
  Serial.begin(9600);
  delay(1500);
  initProperties();
  ArduinoCloud.begin(ArduinoIoTPreferredConnection);
  setDebugMessageLevel(2);
  ArduinoCloud.printDebugInfo();

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(PUMP_PIN, OUTPUT);
  digitalWrite(PUMP_PIN, LOW);

  Wire.begin(SDA_PIN, SCL_PIN);

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED not found");
    while (true);
  }

  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.setTextSize(1);
  display.setCursor(0, 0);
  display.print("System Initialization...");
  display.display();
  delay(2000);

  ThingSpeak.begin(client);
}
```

Software – Loop

```
void loop() {
  ArduinoCloud.update();

  static float lastStableDistance = MAX_HEIGHT;

  unsigned long currentMillis = millis();

  // Measure distance only if pump is OFF
  if (!pumpIsOn) {
    float newDistance = measureDistance();
    if (newDistance > MAX_HEIGHT) newDistance = MAX_HEIGHT;
    if (newDistance >= 0 && newDistance <= MAX_HEIGHT) {
      lastStableDistance = newDistance;
    }
  }

  float waterHeight = MAX_HEIGHT - lastStableDistance;
  if (waterHeight < 0) waterHeight = 0;

  float radius = CYLINDER_DIAMETER / 2.0;
  float volume = 3.1416 * radius * radius * waterHeight;
  float volumeLiters = volume / 1000.0;

  // Pump control
  if (pumpIsOn && currentMillis - lastPumpToggleTime >= pumpOnDuration) {
    digitalWrite(PUMP_PIN, LOW);
    pumpIsOn = false;
    lastPumpToggleTime = currentMillis;
```

Software – Loop

```
    } else if (!pumpIsOn && currentMillis - lastPumpToggleTime >= pumpOffDuration) {
      digitalWrite(PUMP_PIN, HIGH);
      pumpIsOn = true;
      lastPumpToggleTime = currentMillis;
    }

    // ThingSpeak upload
    if (currentMillis - lastUpdate >= updateInterval) {
      lastUpdate = currentMillis;
      ThingSpeak.setField(1, volumeLiters);
      ThingSpeak.setField(2, volumeLiters);
      int x = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
      Serial.println(x == 200 ? "ThingSpeak update successful." : "ThingSpeak error: " + String(x));
    }

    // OLED Display
    display.clearDisplay();
    display.setCursor(0, 0);
    display.setTextSize(1);
    display.print("Water Height: ");
    display.print(waterHeight, 1);
    display.println(" cm");

    display.print("Volume: ");
    display.print(volumeLiters, 2);
    display.println(" L");

    display.print("Pump: ");
    display.println(pumpIsOn ? "ON" : "OFF");

    display.display();

    delay(300);
  }
```

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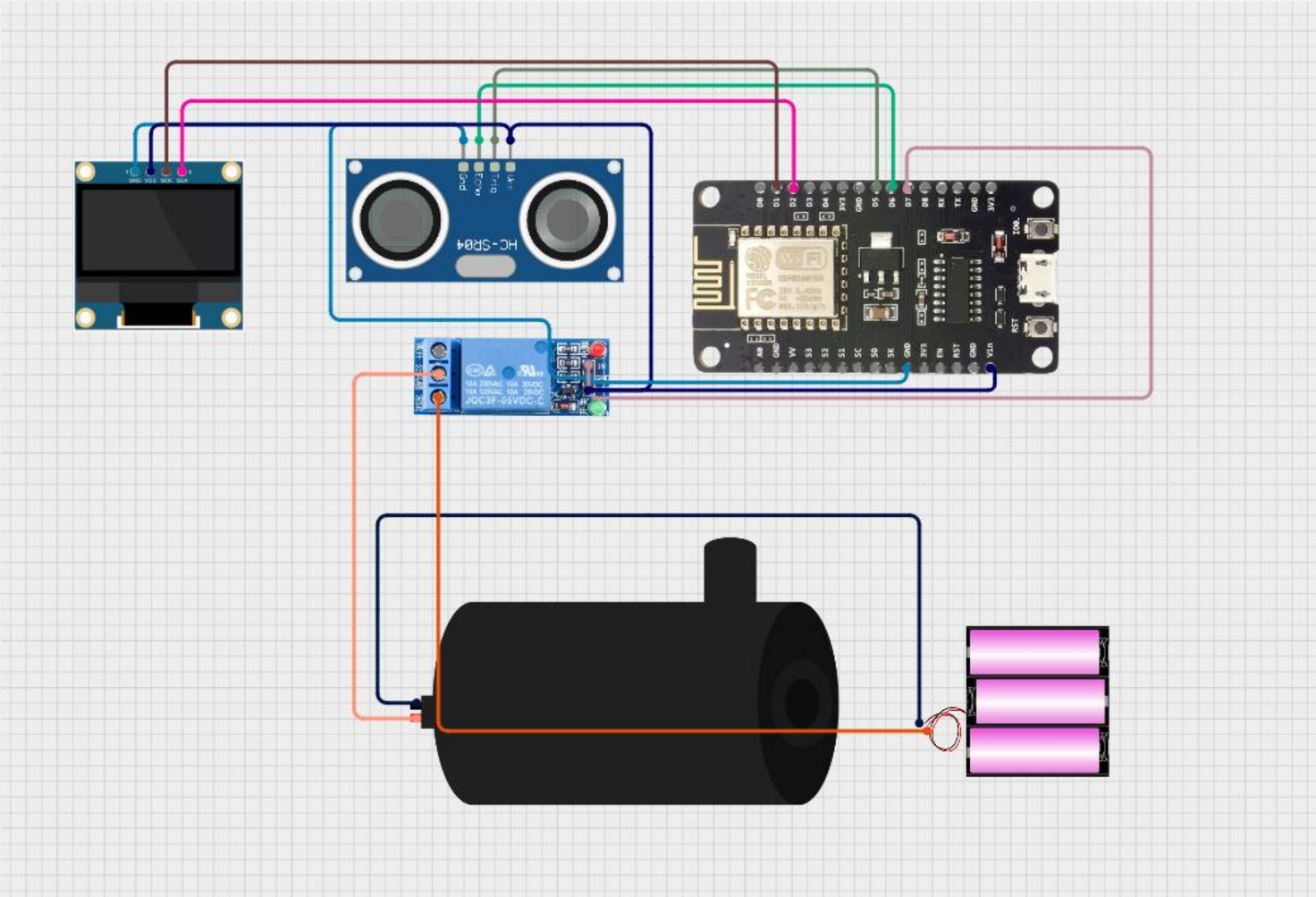
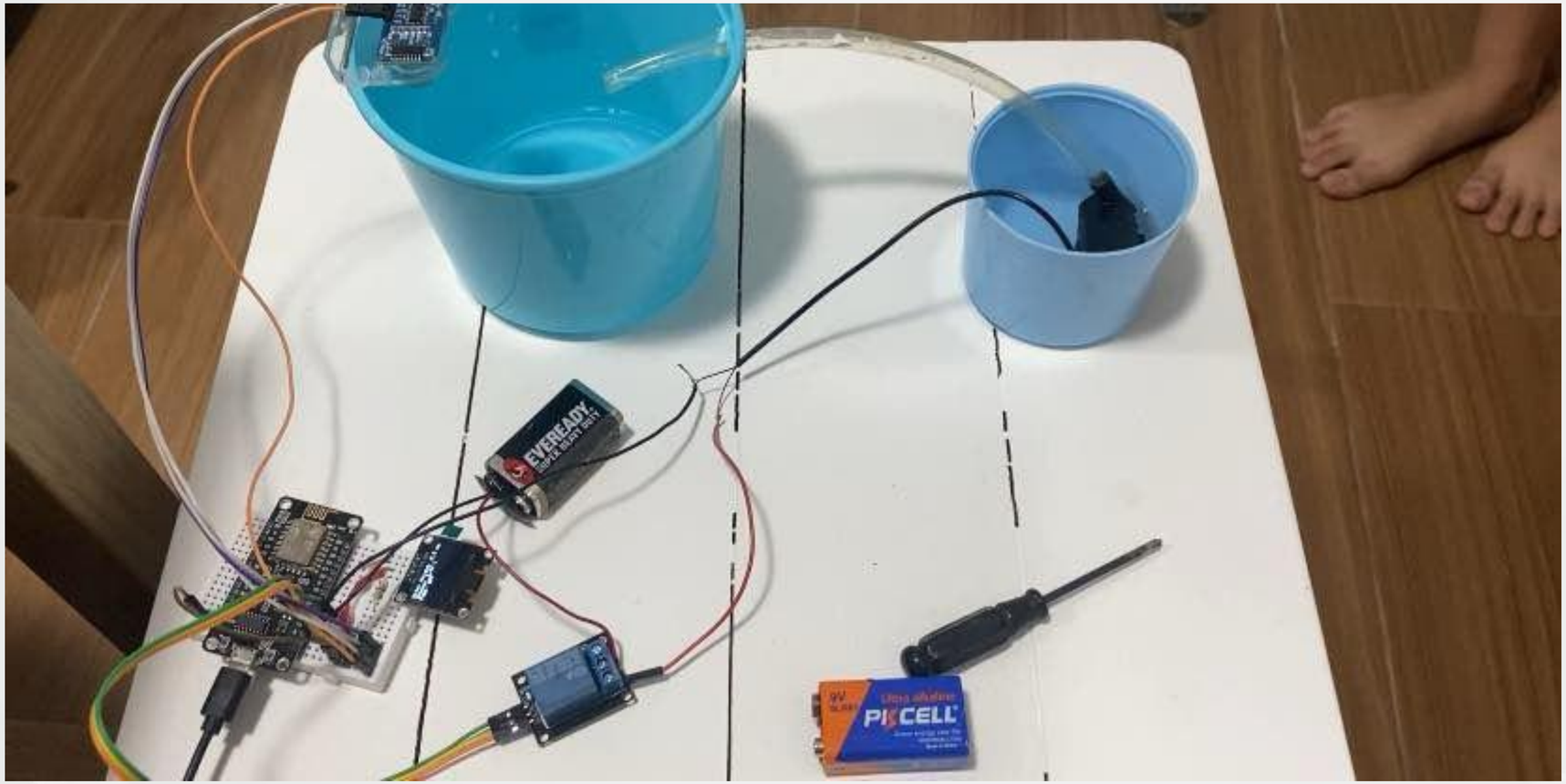
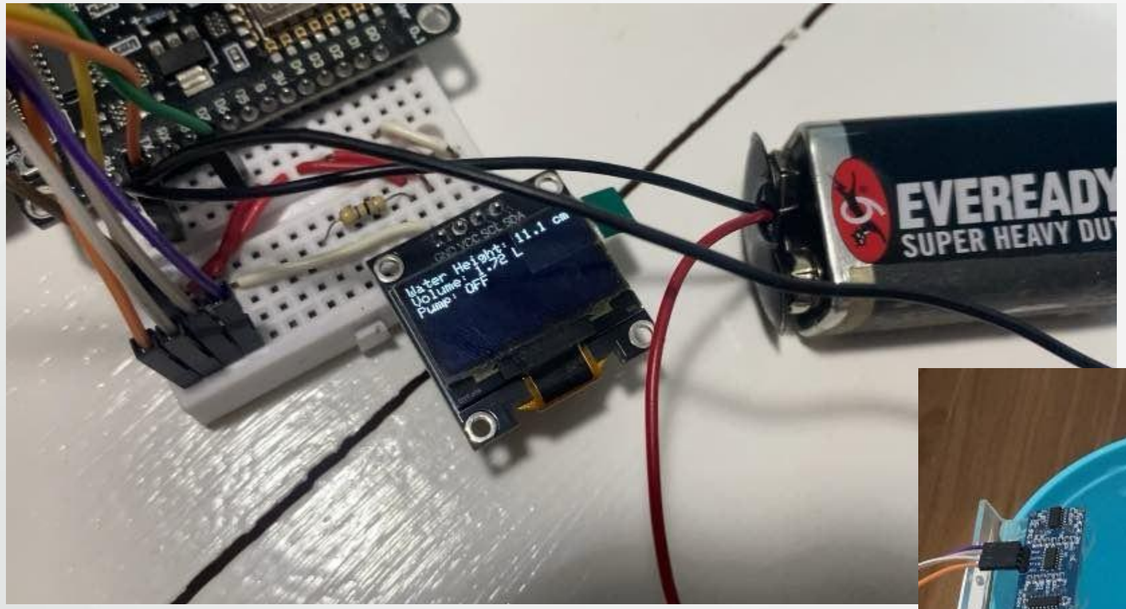
Hardware – Circuit Diagram

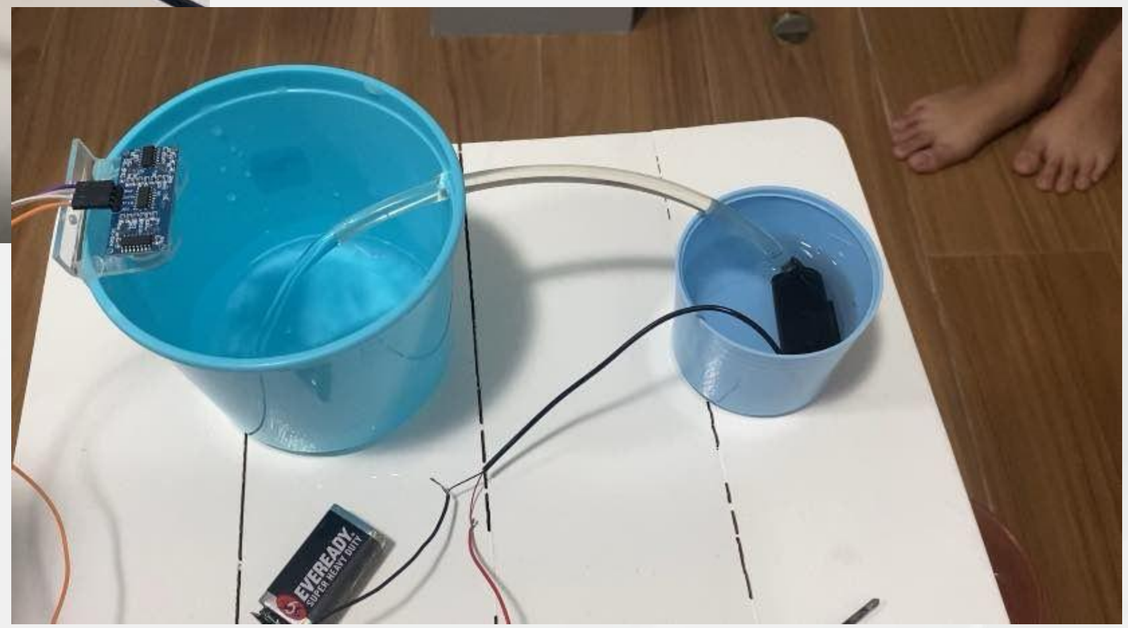


Hardware – Circuit Diagram (Actual Implimentation)



Sample Output





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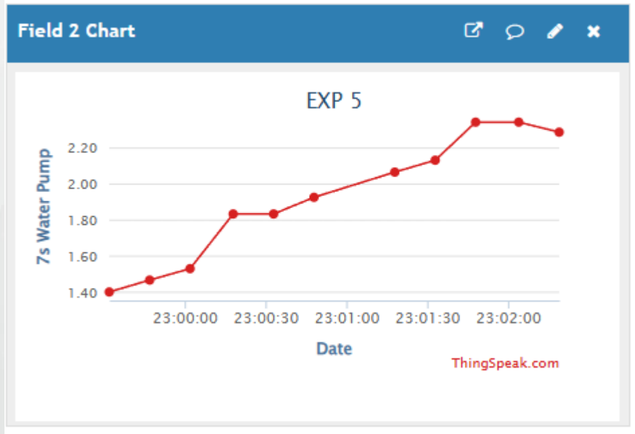

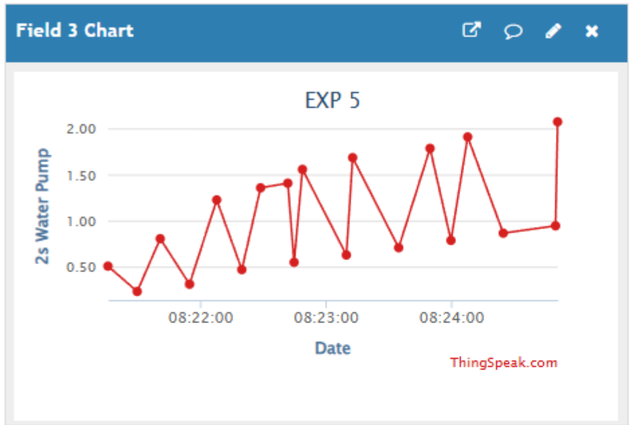

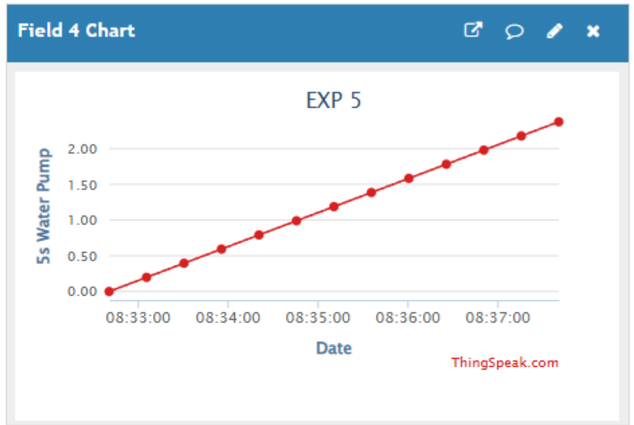

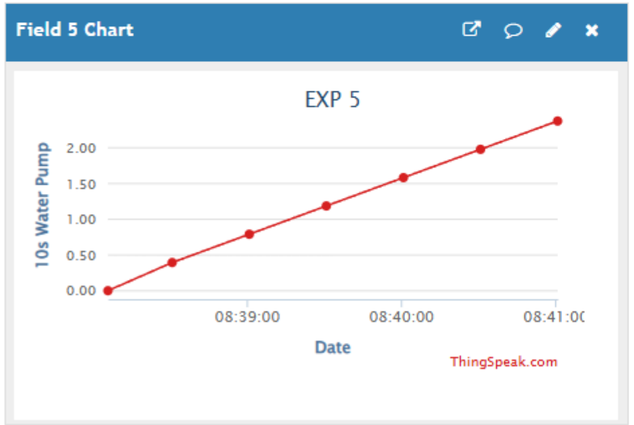

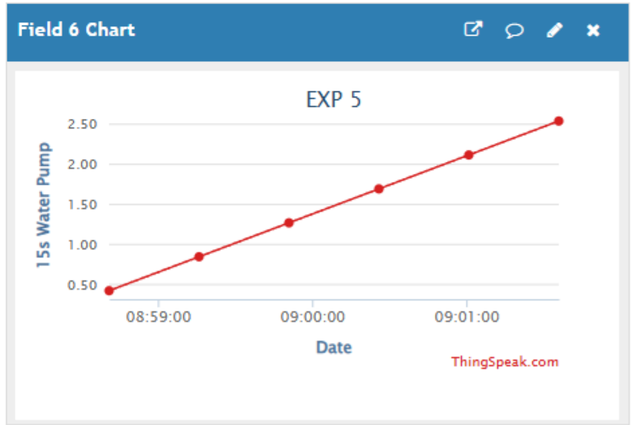

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DEMONSTRATION

Link: https://youtu.be/dtTNeBUxfNk

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